SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4307

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I _G. Srivalli (192365104)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Srivalli

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Question 1: Multimedia Integration

Select the correct HTML tag for video embedding and justify your answer.

Selected Option

B. `<video src="video.mp4" controls></video>`

Detailed Justification

- **HTML5 Native Capabilities:** HTML5 introduced the native `<video>` element, allowing browsers to render media files directly using built-in decoding engines without relying on external plugins, active controls, or proprietary software.
- **Deprecation of Legacy Plugins:** Legacy implementations relied on third-party browser plugins like Adobe Flash or Silverlight, which suffered from severe security vulnerabilities, high CPU overhead, lack of mobile support, and memory leaks. Modern standards require native rendering for stability and cross-platform compatibility.
- **Role of the `controls` Attribute:** The boolean `controls` attribute instructs the user agent to display the browser's default playback interface—including Play/Pause, volume slider, seek bar, and fullscreen toggles—ensuring standard user accessibility.
- **Analysis of Invalid Options:**
    - **Option A (`<media>`):** Non-standard element; does not exist in W3C HTML specifications.
    - **Option C (`<movie>`):** Proprietary or non-standard tag; unrecognized by rendering engines.
    - **Option D (`<vid>`):** Invalid abbreviation; unrecognized by standard HTML parsers.

Question 2: Secure Form Submission

Identify the appropriate HTTP method for form submission and explain why.

Selected Option

B. `POST`

Detailed Justification

- **Data Transmission Mechanism:** The `POST` method encapsulates sensitive payload data within the body of the HTTP request packet rather than appending key-value pairs directly to the Uniform Resource Identifier (URI).
- **Vulnerabilities of the `GET` Method:**
    - **URL Visibility:** `GET` appends form parameters as query strings (`?user=admin&pass=secret`), exposing sensitive user data on screens and displays.
    - **Browser History & Logs:** Query strings are recorded in browser history, proxy logs, CDN traces, and server access logs (`access.log`), leading to credential leaks.
    - **Length Limits:** Browsers and web servers impose URI length restrictions (typically 2,048 to 8,192 characters), making `GET` unsuitable for large form payloads.
- **Inappropriateness of `PUT` and `DELETE`:**
    - `PUT` is designed for resource replacement at a specific URI endpoint, creating or updating an exact target entity.
    - `DELETE` is intended for explicit resource removal from a target server location.
    - Neither `PUT` nor `DELETE` are directly supported as native HTML form `method` attributes without custom AJAX/Fetch middleware wrappers.

Question 3: Semantic Architecture & Accessibility

Choose the proper semantic element for navigation and justify its usage.

Selected Option

B. <nav>

Detailed Justification

- **Semantic Meaning in HTML5:** The <nav> element explicitly designates a section of a document containing primary navigation links, establishing structural intent within the Document Object Model (DOM).
- **Accessibility & Assistive Technologies:** Screen readers (e.g., NVDA, JAWS, VoiceOver) use landmark roles to parse documents. The <nav> tag automatically exposes a landmark region, allowing users with visual impairments to jump straight to main site navigation.
- **Search Engine Optimization (SEO):** Search engine web crawlers evaluate semantic tags to prioritize site architecture mapping, indexing core navigation hierarchies more efficiently.
- **Comparison with Non-Semantic Containers:**
    - ** :** A completely non-semantic container with zero structural meaning. It provides no context to search engines or screen readers.
    - **<section>:** Represents a generic document section (e.g., chapters, news feeds); lacking specific navigation semantics.
    - **<header>:** Intended for introductory content, page logos, or heading groups; not designed exclusively as a main link container.

Question 4: XML Syntax & Validation

Identify the correct XML syntax and explain the error in other options.

Selected Option

B. `<note><to>User</to><from>Admin</from></note>`

Detailed Structural Analysis

Option	Syntax Sample	Structural Status	Analysis & Error Identification
A	`<note><to>User</to><from>Admin</from>`	**Invalid**	**Missing Closing Root Tag:** The root element `<note>` is opened but never closed with `</note>`, violating fundamental XML well-formedness standards.
B	`<note><to>User</to><from>Admin</from></note>`	**Valid**	**Well-Formed XML:** Strictly complies with all XML specifications: contains a single root element, proper opening/closing tags, and correct element nesting.
C	`<note><to>User</from></note>`	**Invalid**	**Tag Mismatch / Bad Nesting:** The element opened as `<to>` is improperly closed with `</from>`. Element tags must match case-sensitively and close in sequence.
D	`<note to="User"></note`	**Invalid**	**Malformed Syntax:** The closing bracket > is missing from the terminal tag `</note`, causing a fatal parsing syntax error.

Question 5: Application Protocols

Choose the correct protocol for web communication and justify your choice.

Selected Option

B. HTTP (Hypertext Transfer Protocol)

Detailed Justification

- **Web Native Design:** HTTP is an application-layer request-response protocol running over TCP/IP, specifically designed for transferring hypermedia resources (HTML, CSS, JS, images, media) across distributed networks.
- **The Role of HTTPS:** Modern web deployments enforce HTTPS (HTTP over TLS/SSL) to protect data confidentiality, preserve integrity via encryption, and verify server identities using public key infrastructure.
- **Inappropriateness of Alternative Protocols:**
    - **FTP (File Transfer Protocol):** Built for remote file transfer and management, lacking native web browser rendering mechanics or stateless request-response workflows.
    - **SMTP (Simple Mail Transfer Protocol):** Designed for email payload routing and delivery between mail transfer agents (MTAs), not interactive web page content delivery.
    - **SNMP (Simple Network Management Protocol):** Used for monitoring network appliances (routers, switches, servers), totally detached from content transport.